

EE-102 Electric Circuit Analysis

Experiment # 5 & 6

To Verify Kirchoff's Voltage and Current Laws

Objective/s:

- To verify Kirchoff's voltage and current laws

Components

- Resistors
- DMM
- DC power supply

1. Theory

Consider the simple series circuit Fig. Here we have numbered the points in the

Circuit for voltage reference. As we are dealing with dc circuits, therefore we should carefully connect the voltmeter while measuring voltage across supply or any of the resistances as shown in fig, keeping in mind the similarity of polarities of voltage across the element and that of the connected probes of meter. In such case, we will observe that,

$$\begin{array}{ll}
 E_{2-1} = +45 \text{ V} & \text{voltage from point 2 to point 1} \\
 E_{3-2} = -10 \text{ V} & \text{voltage from point 3 to point 2} \\
 E_{4-3} = -20 \text{ V} & \text{voltage from point 4 to point 3} \\
 + E_{1-4} = -15 \text{ V} & \text{voltage from point 1 to point 4} \\
 \hline
 & 0 \text{ V}
 \end{array}$$

This principle is known as **Kerchief's Voltage Law**, and it can be stated as such:

"The algebraic sum of all voltages in a loop must equal zero"

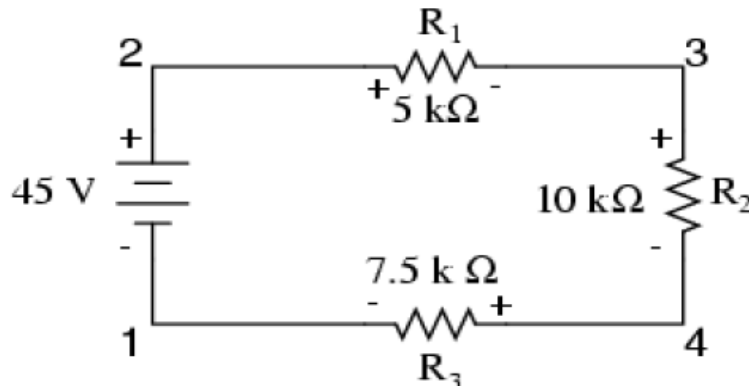


Figure 1

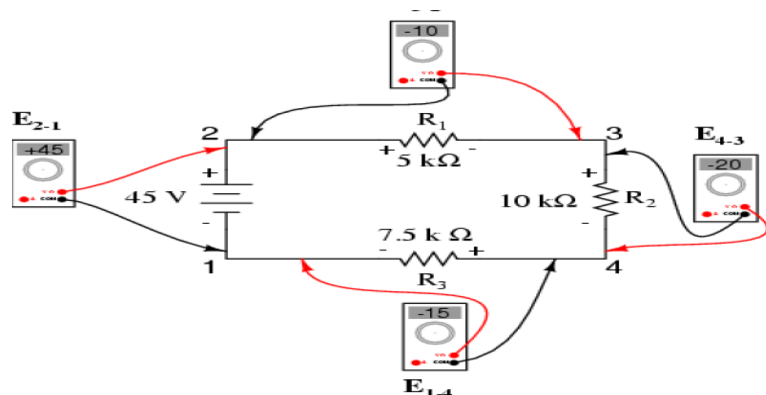


Figure 2

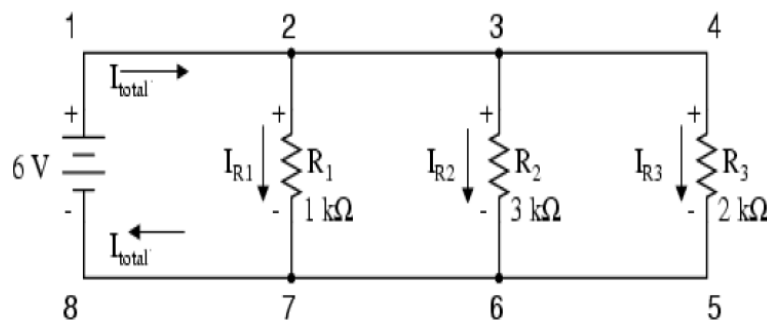


Figure 3

Let's take a closer look at the circuit given in Fig 3: Then, according to Kirchhoff's Current Law: **"The algebraic sum of all currents entering and exiting a node must equal zero"** Mathematically, we can express this general relationship as such:

That is, if we assign a mathematical sign (polarity) to each current, denoting whether they Enter (+) or exit (-) a node, we can add them together to arrive at a total of zero, guaranteed.

2. Tasks

2.1 Kirchhoff's Voltage Law

1. Construct circuit of Fig. 1 using the values R_1 , R_2 , R_3 as shown in the figure.
2. Adjust the output of the power supply so that $V = 40V$. Measure and record this Voltage in table 1, also measure and record the voltages V_1 , V_2 , V_3 and Enter the sum in the same table.

2.1.1 Observations

V_T	V_1	V_2	V_3	Sum

Table 1

2.2 Kirchhoff's Current Law

1. Connect the circuit of Fig. 3 with $V = 40 V$.
2. Measure and record in Table 2 currents I_{R1} , I_{R2} , I_{R3} and I_{total} .

3 Observations

I_{Total}	I_{R1}	I_{R2}	I_{R3}	Sum

Table 2